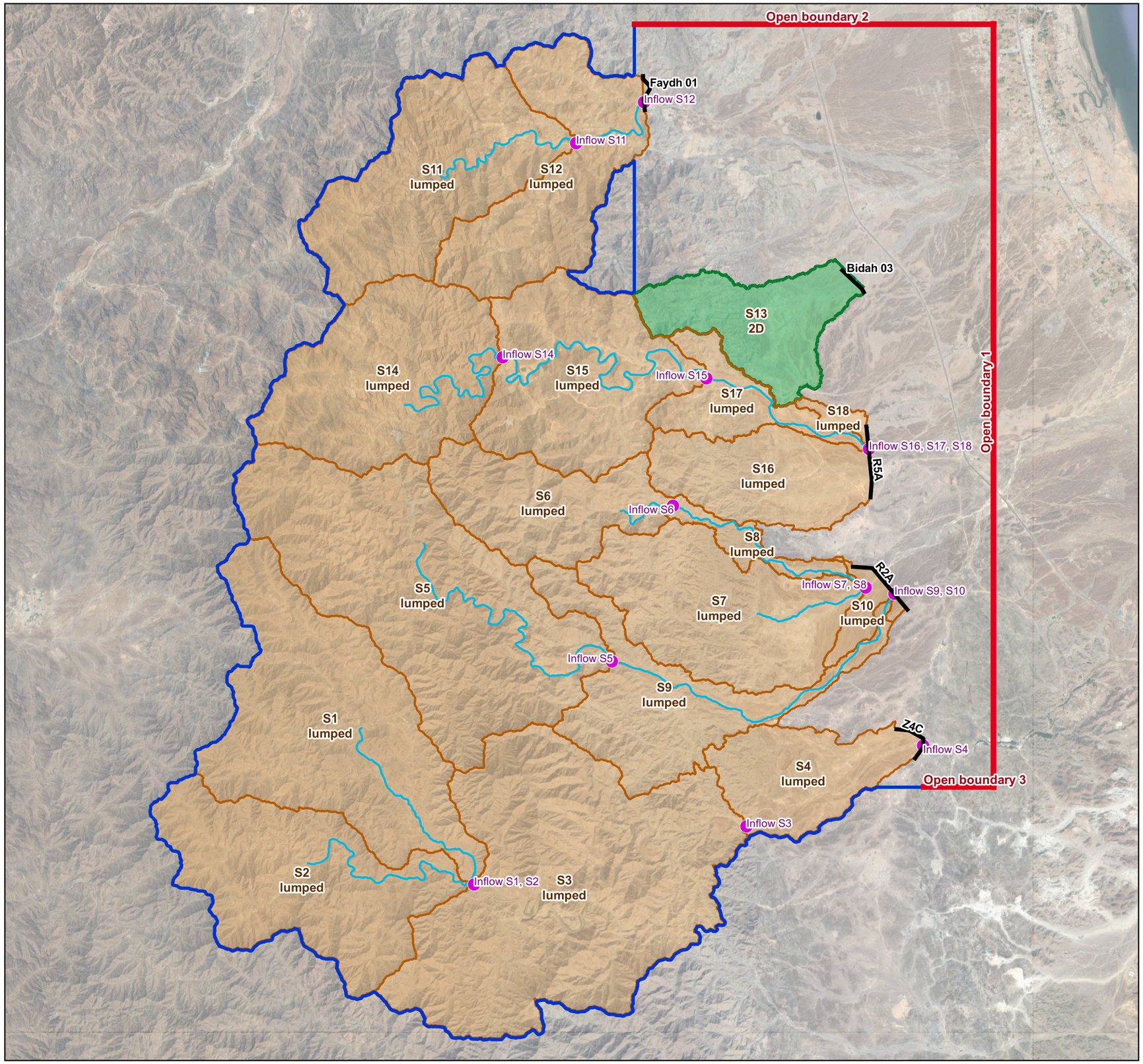
Wadi2D hybrid model, Block 2: setup

How the model is built



Legend

Subbasins

- Subbasin, lumped (rain masked in 2D)
- Subbasin, 2D rain-on-grid
- 2D model domain, closed walls except open boundaries
- Open boundary
- Lumped hydrograph enters 2D here
- Reach, routed by the 2D model here
- Design line

Hills: 17 subbasins run in the lumped engine. Their rain is masked out of the 2D model and each hydrograph enters at its magenta point.

2D: S13 and the plain, rain on the grid, 20 m cells with terrain tables from the DEM, Manning n 0.04.

Water leaves through 3 open boundaries. Every other edge is a closed wall.

Storm: 251 mm, storm area 300 km2. Simulated 48 h.

Water: 103.3 Mm3 from the lumped hills, 39.1 Mm3 of rain on the 2D area after losses, 139.0 Mm3 out. Mass error 2e-08.

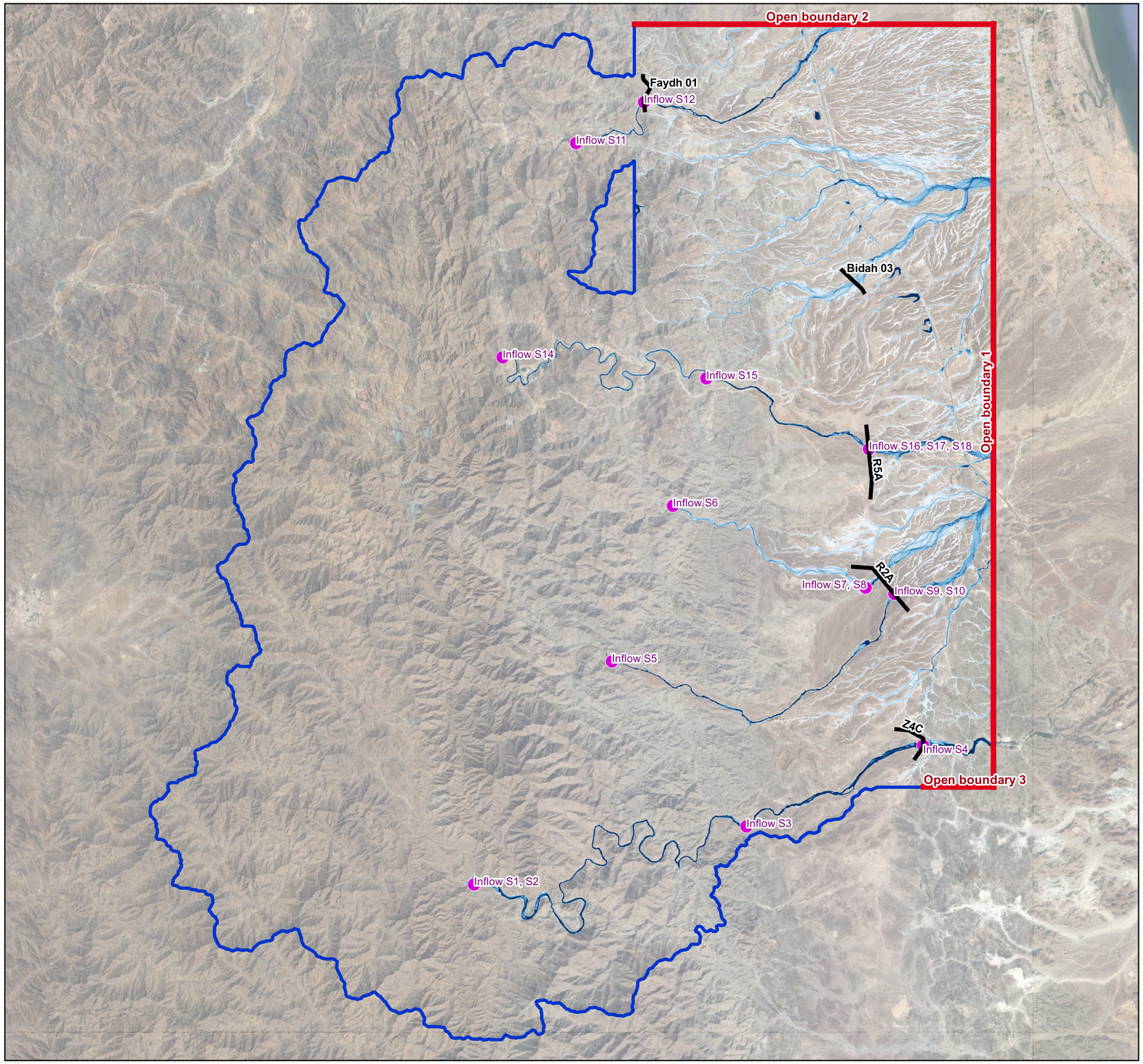
Not a design result. No gauge, no culverts, 20 m cells.

0 2.5 5 7.5 10 km

EPSG:32640. Imagery: Google.

Wadi2D hybrid model, Block 2: maximum depth

Depth at the 5 m pixel, sloped water surface



Legend

Maximum depth, m

- 3 m
- 0 m
- 2D model domain, closed walls except open boundaries
- Open boundary
- Lumped hydrograph enters 2D here
- Design line

Peak flow at the design lines

Z4C: 2,040 m3/s at hour 14.9
R2A: 1,545 m3/s at hour 13.7
R5A: 1,030 m3/s at hour 14.1
Faydh 01: 849 m3/s at hour 13.3
Bidah 03: 668 m3/s at hour 12.3

Water: 103.3 Mm3 from the lumped hills, 39.1 Mm3 of rain on the 2D area after losses, 139.0 Mm3 out. Mass error 2e-08.

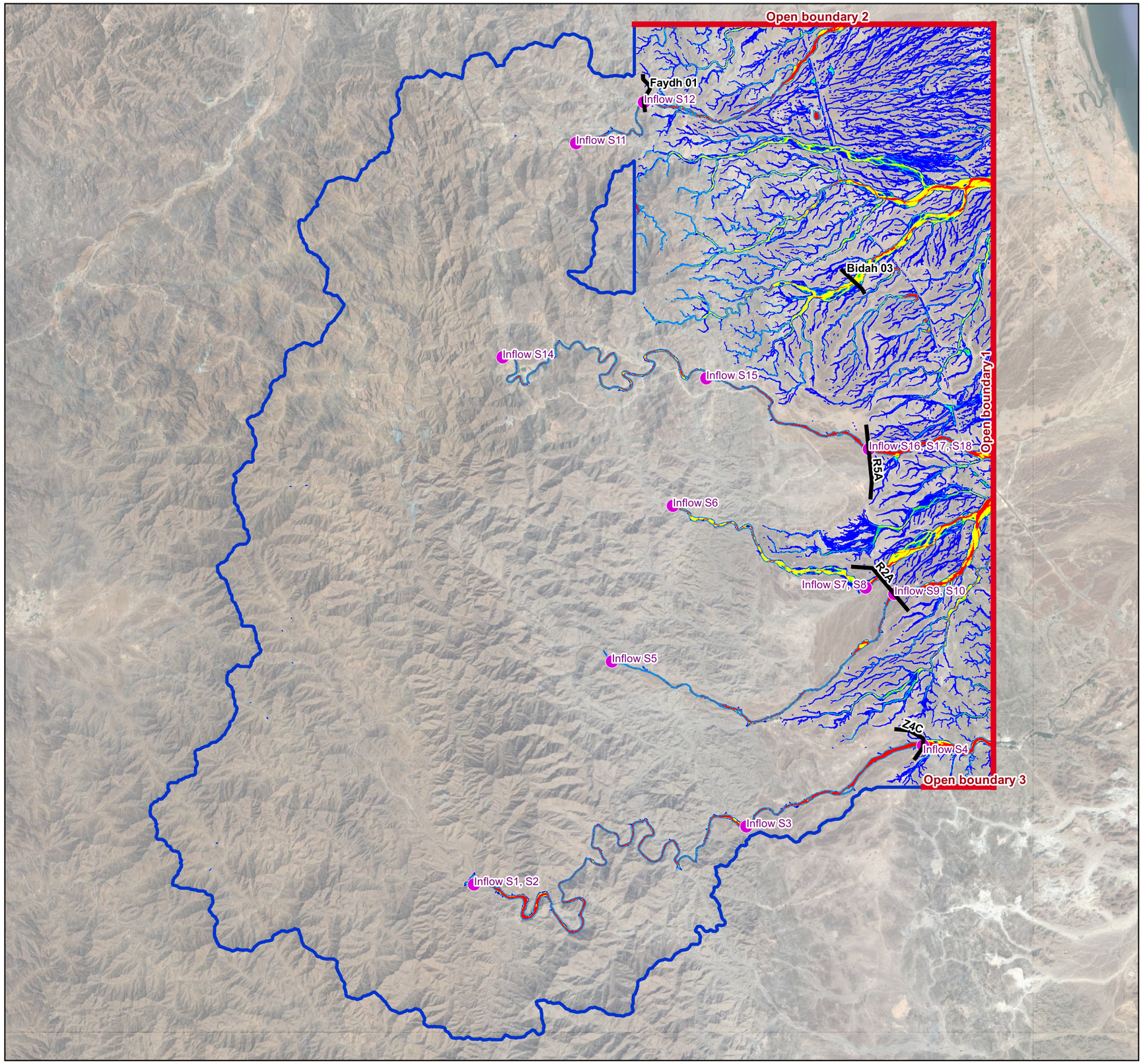
Not a design result. No gauge, no culverts, 20 m cells.

0 2.5 5 7.5 10 km

EPSG:32640. Imagery: Google.

Wadi2D hybrid model, Block 2: AIDR hazard H1 to H6

Pixel depth x interpolated velocity, 15-minute maps



Legend

AIDR hazard class

- H1 D x V 0.3, D 0.3, V 2
- H2 D x V 0.6, D 0.5, V 2
- H3 D x V 0.6, D 1.2, V 2
- H4 D x V 1, D 2, V 2
- H5 D x V 4, D 4, V 4
- H6 above H5
- 2D model domain, closed walls except open boundaries
- Open boundary
- Lumped hydrograph enters 2D here
- Design line

Area by class, 51.0 km2 in all

H1: 16.6 km2
H2: 6.8 km2
H3: 1.0 km2
H4: 4.0 km2
H5: 12.3 km2
H6: 10.3 km2

A pixel takes the highest class any limit gives. The legend gives each class's upper limits: D x V in m2/s, D in m, V in m/s. Depth x speed at the same moment, from the saved maps, with the run's own peaks between them.

Not a design result. No gauge, no culverts, 20 m cells.

0 2.5 5 7.5 10 km

EPSG:32640. Imagery: Google.